# DTIE Presentation Project

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## Overview

This DTIE project proposes an AI crop recommender that turns a small farmer's own soil report and local weather into a best-crop suggestion, with reasons, in the farmer's language. It targets the guesswork behind crop choice, which causes avoidable yield and income loss every season.

| Item | Detail |
|---|---|
| Title | AI Crop-Based Recommendation System for Farmers |
| Course | Design Thinking for Innovation & Entrepreneurship (DTIE) |
| Supervisor | Dr Sudhakar |
| Team | Jaswin V (2620030148), Ganesh N (2620030158), Navadeep (2620030007), Rohan (2620030090), Revanth (2620030100), Ritwik (2620030220) |
| Method | Design thinking: problem analysis, empathy, define, ideate, vote, concept, prototype, demo |

## Problem analysis

Crop choice stays a guess because six factors work against the farmer at once. The problem statement: *crop selection is a guess, so avoidable yield and income loss repeats every season.*

| Lens | Finding |
|---|---|
| Man | Small and marginal farmers pick crops from habit, family tradition or a neighbour's advice, not their own soil data. |

| Lens | Finding |
|---|---|
| Machine | No affordable, local-language advisory tool; the Soil Health Card portal gives fertiliser doses, not a crop choice. |
| Material | Soil is tested once in 2–3 years on a limited nutrient set; rainfall, humidity and temperature are never combined with it. |
| Method | Crop is picked before sowing, and soil testing rarely feeds back into the next season's decision. |
| Measurement | No plot-level yield baseline, so a wrong crop is discovered only at harvest. |
| Mother Nature | Erratic monsoon and rising temperatures keep shifting the safe sowing window. |

## Empathy

The farmer does not need more data; they need one clear, trustworthy, local-language answer before sowing.

| | Ramesh (42, 2 acres) | Kavita (33, 1 acre) |
|---|---|---|
| Profile | Rain-fed holding, smartphone user, speaks Marathi; banks on last year's crop when the new season begins. | Grows vegetables; wants a second-income crop but does not know what her soil supports. |

Empathy map:

- **Says:** "I sow what my father sowed, what everyone in the village is sowing." / "The officer's advice is the same for the whole village; my plot is different."
- **Does:** Visits the input dealer first, buys seed on credit, tests soil only if the card is free.
- **Feels:** Anxious after a loss, distrustful of apps in English, but proud when a season goes well.

# Define

**Point of view:** a rain-fed small farmer needs a simple, credible crop suggestion built from their own soil and local weather, because one-size-for-the-village advice turns sowing into a gamble.

How might we...

1. make a soil-test result instantly useful, as an answer and not a chart?
2. give the suggestion in the farmer's own language, offline, on a low-end phone?
3. show why a crop was recommended so the farmer can trust it?
4. warn early when the weather turns against the chosen crop?

**Design criteria:** accuracy ≥ 90%, answer in under 5 seconds, works offline, 3 taps or fewer, local language, explains top factors, free for the farmer.

# Idea generation and voting

The team listed twelve starting points, from a soil and weather model to farmer-to-farmer learning. Idea 1, the ML crop recommender, scored highest (5 on a 1–5 scale) in group discussion and dot voting and was carried forward.

| # | Idea |
|---|---|
| 1 | ML crop recommender from soil + weather (N, P, K, pH, rainfall, temperature, humidity) |
| 2 | SMS / IVR advisory for feature-phone farmers |
| 3 | Village-level crop board updated by the extension officer |
| 4 | Low-cost soil sensor kit bundled with the app |
| 5 | Voice input in local language |
| 6 | Crop + profit calculator per acre before sowing |
| 7 | Rotational-crop planner across 3 seasons |
| 8 | Weather-linked sowing-date alert |
| 9 | Image-based leaf disease check after sowing |

| # | Idea |
|---|---|
| 10 | Farmer-to-farmer advisory from past seasons |
| 11 | Fertiliser and irrigation plan for the chosen crop |
| 12 | Mandi price and nearest-market link |

## Solution concept and rapid estimate

The concept is a model-driven recommender that takes the farmer's soil report and local weather and returns the most suitable crop with the top reasons, in the local language.

- **Inputs:** N, P, K, soil pH, organic carbon, rainfall, temperature, humidity, soil type, season.
- **Output:** best crop, top-3 alternatives, confidence %, top influencing factors and a next step.
- **Architecture:** farmer app / SMS → REST API → trained ML model → soil + weather database → advisory.
- **Rapid estimate:** 2,200+ records, 22 crops, 7 features; 4–6 weeks with a 3-member team.
- **Feasibility:** free and open-source tooling (Python, scikit-learn, Flask); cloud free tier plus an offline model on the device.
- **Risks and mitigation:** bad soil input → range validation; farmer distrust → explainable factors; poor network → offline mode.

## Rough prototype

The prototype has three screens: enter soil values (type them, or enter the Soil Health Card number), a recommendation card in the local language, and a "Why this crop?" view comparing the pick with runner-up crops.

The model was built on a public crop-recommendation dataset of 2,200 rows and 22 crops, enriched with local KVK plot records. Five algorithms were compared with 5-fold cross-validation and an 80:20 train/test split: Random Forest, XGBoost, Naive Bayes, SVM and KNN.

Published benchmarks for comparison: Random Forest 98.86% (Scientific Reports, 2025); stacked ensembles 98.4–99.54%; RF 99.09% vs Naive Bayes 90.9% (arXiv, 2025).

## Demonstration and validation

The demo shows a farmer entering a real soil report, getting the crop with a confidence score, opening "Why?", then changing rainfall to show the recommendation flip.

- **Expected results:** test-set accuracy of at least 90% (target 95%+ with Random Forest) and latency under 5 seconds.
- **Validation:** k-fold cross-validation, a confusion matrix, and a blind check of 15 farmer plots against the crop actually grown.
- **Farmer feedback:** 5 farmers score the suggestion for clarity, language and trustworthiness.
- **Impact:** weed and management-related losses alone are put at up to 31.5% in Indian field studies, so a better crop and input choice reduces avoidable yield loss.
- **Future scope:** market price and profit forecast, pest and disease advisory, weather alerts, more crops and regional languages.

## Conclusion

AI will not replace the Indian farmer; it will stand beside them.

- **Better decisions:** AI turns local data into simple, on-time advice.
- **Higher productivity:** the right crop and the right input give a measurable yield uplift.
- **Sustainable farming:** less water, less blanket fertiliser, healthier soil.
- **Next steps:** drones, IoT sensors and disease imaging.

## Key references

Pudumalar et al., IEEE IC3 2017; Mohapatra & Kale, ITEGAM-JETIA 2024; Shastri et al., Scientific Reports 2025; Gharde et al., 2018. Taken from the presentation's reference slide; not independently re-checked.